Original Lie algebra mini lesson

A bracket can be computed as [X,Y] = XY - YX.

X = diag(1,-1), Y has Y_12 = 1 and other entries zero.

XY = Y and YX = -Y. Therefore [X,Y] = 2Y.

Exercise: compute [X,3Y] by the same two products.

Original analysis mini lesson

Consider f(x) = x^2 and a fixed real number a.

f(a+h)-f(a) = 2ah+h^2.

For 0 < |h| < 1, |2ah+h^2| <= (2|a|+1)|h|.

Choose delta = min(1, epsilon/(2|a|+1)).

Then |h| < delta implies |f(a+h)-f(a)| < epsilon.

Original public test material - no private course content